

APK Analysis of instagram_lite_v510.0.0.10.323.apk (Instagram Lite)

This report presents a static security analysis of **instagram_lite_v510.0.0.10.323.apk**
(Instagram Lite) by

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Executive Summary

This report presents the findings of a static security analysis conducted on Instagram Lite (v510.0.0.10.323), an Android application developed by Meta Platforms. The analysis was performed using MobSF for automated static analysis, VirusTotal for malware and hash verification, and Exodus Privacy for tracker and permission profiling.

MobSF assigned the application a security score of 53/100 with a Risk Rating of Medium, identifying a total of 3 high-severity, 57 medium-severity, 2 informational, 5 secure, and 1 hotspot finding. The 3 high-severity findings are as follows: (1) the base network security configuration permits cleartext HTTP traffic to all domains globally; (2) domain-specific cleartext traffic is permitted to h.facebook.com, l.facebook.com, l.alpha.facebook.com, lm.facebook.com, and l.instagram.com; and (3) the application supports installation on Android 5.0–5.0.2 (minSdk=21), which is an unpatched and vulnerable platform version.

Source code analysis further identified insecure use of java.util.Random (CWE-330, MSTG-CRYPTO-6), world-readable external storage access in X/C01V.java (CWE-276, MSTG-STORAGE-2), MD5 hashing in X/C0BO.java (CWE-327, MSTG-CRYPTO-4), SHA-1 hashing in X/AnonymousClass041.java (CWE-327, MSTG-CRYPTO-4), and hardcoded IP address disclosure in X/C00A.java (CWE-200, MSTG-CODE-2).

VirusTotal analysis of the APK hash (SHA256: 3e3453faede38edb7902a59df3f8ee34f5e397bad99b5663362d3af7498a5718) returned a community score of 0/67 — no security vendors flagged the file as malicious. Exodus Privacy identified 0 trackers and 61 total permissions. Application permissions analysis revealed 22 dangerous permissions, of which 16 out of 25 match known top malware permission patterns, including CAMERA, RECORD_AUDIO, SYSTEM_ALERT_WINDOW, READ_CONTACTS, and READ_PHONE_STATE.

Component exposure analysis shows 10 of 21 activities, 3 of 44 services, 15 of 43 receivers, and 8 of 13 providers are exported — representing a significant attack surface for unauthorized inter-app interaction.

An OWASP MASVS checklist review confirmed failure in 6 out of 9 security areas. Overall, the application presents a medium-risk security posture with critical weaknesses in network transport security, cryptographic implementation, and permission management.

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Chapter 1 — Objective

The objective of this assessment is to perform a static security analysis of the **Instagram Lite** Android application (v 510.0.0.10.323) using the Mobile Security Framework (MobSF). This assessment aims to identify security vulnerabilities, evaluate dangerous permissions and exported components, analyze network security configurations, and assess the overall security posture of the application. The findings are cross-referenced with VirusTotal and Exodus Privacy to provide a comprehensive security evaluation.

Chapter 2 — Methodology

1. List of used Tools & Utility

MobSF - Mobile Security Framework (Local offline)
VirusTotal (Online)
Exodus Privacy

2. Step of Analysis

1. Download **Instagram_lite_v510.0.0.10.323.apk** From
<https://www.instagram.com/web/lite/>

2. This **Instagram_lite_v510.0.0.10.323.apk** Upload to MobSF (Local Machine) for static analysis. Run static analysis by MobSF (Local Machine) for-

1. Application Overview
2. App Components
3. Security Score Analysis
4. Application Permissions
5. Identified Problems — Finding (High)

3. Malicious analysis from **virustotal.com** by using **MobSF Reported hash value** for this apk.

4. Found **Tracker & Permission** used by **Exodus Privacy** (from <https://reports.exodus-privacy.eu.org/en/>)

5. Preparing **Manual Checklist** based on **OWASP Mobile Security Guidelines**

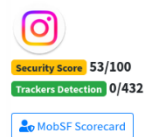
6. Source Code Review

Chapter 3 : Work Details

1. Application Overview

App Name	Instagram Lite
Package Name	com.instagram.lite
Main Activity	com.facebook.lite.MainActivity
Android Version Name	510.0.0.10.323
Android Version Code	959256971
File Name	instagram_lite_v510.0.0.10.323.apk
Size	3.53MB
MD5	f7adec6949c198e387441946be163d96
SHA1	7abe0829d1b910fd0282df9687eed4952c208cc7
SHA256	3e3453faede38edb7902a59df3f8ee34f5e397bad99b5663362d3af7498a5718

APP SCORES



FILE INFORMATION

File Name	instagram_lite_v510.0.0.10.323.apk
Size	3.53MB
MD5	f7adec6949c198e387441946be163d96
SHA1	7abe0829d1b910fd0282df9687eed4952c208cc7
SHA256	3e3453faede38edb7902a59df3f8ee34f5e397bad99b5663362d3af7498a5718

APP INFORMATION

App Name	Instagram Lite		
Package Name	com.instagram.lite		
Main Activity	com.facebook.lite.MainActivity		
Target SDK	36	Min SDK	21
	Max SDK		
Android Version Name	510.0.0.10.323	Android Version Code	959256971

2. App Components

EXPORTED ACTIVITIES 10 / 21

EXPORTED SERVICES 3 / 44

EXPORTED RECEIVERS 15 / 43

EXPORTED PROVIDERS 8 / 13

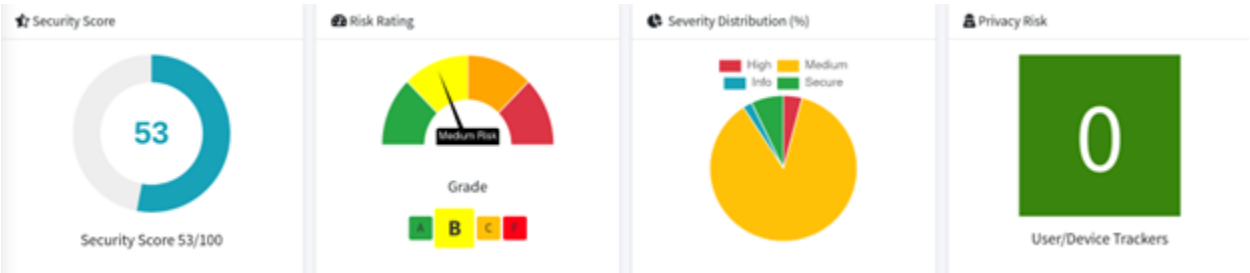


3. Security Score Analysis

Security Score53/100

Trackers Detection0/432

Risk RatingMEDIUM



- Findings
- High3
- Medium57
- Info2
- Secure5
- Hotspot1

Findings

High

3

Medium

57

Info

2

Secure

5

Hotspot

1

4. APPLICATION PERMISSIONS

5. Found 22 critical / Dangerous permission(s)

6.

hotspot

Found 22 critical permission(s)

PERMISSIONS

PERMISSION	STATUS	INFO	DESCRIPTION	CODE MAPPINGS
android.permission.ACCESS_COARSE_LOCATION	Dangerous	coarse (network-based) location	Access coarse location sources, such as the mobile network database, to determine an approximate phone location, where available. Malicious applications can use this to determine approximately where you are.	
android.permission.ACCESS_FINE_LOCATION	Dangerous	fine (GPS) location	Access fine location sources, such as the Global Positioning System on the phone, where available. Malicious applications can use this to determine where you are and may consume additional battery power.	
android.permission.AUTHENTICATE_ACCOUNTS	Dangerous	act as an account authenticator	Allows an application to use the account authenticator capabilities of the Account Manager, including creating accounts as well as obtaining and setting their passwords.	
android.permission.CALL_PHONE	Dangerous	directly call phone numbers	Allows the application to call phone numbers without your intervention. Malicious applications may cause unexpected calls on your phone bill. Note that this does not allow the application to call emergency numbers.	
android.permission.CAMERA	Dangerous	take pictures and videos	Allows application to take pictures and videos with the camera. This allows the application to collect images that the camera is seeing at any time.	
android.permission.GET_ACCOUNTS	Dangerous	list accounts	Allows access to the list of accounts in the Accounts Service.	
android.permission.GET_TASKS	Dangerous	retrieve running applications	Allows application to retrieve information about currently and recently running tasks. May allow malicious applications to discover private information about other applications.	Hotspot
android.permission.MANAGE_ACCOUNTS	Dangerous	manage the accounts list	Allows an application to perform operations like adding and removing accounts and deleting their password.	
android.permission.POST_NOTIFICATIONS	Dangerous	allows an app to post notifications.	Allows an app to post notifications	
android.permission.READ_CALENDAR	Dangerous	read calendar events	Allows an application to read all of the calendar events stored on your phone. Malicious applications can use this to send your calendar events to other people.	Hotspot

Showing 1 to 10 of 66 entries

Previous

1234567

Next

7. Identified Problems — Finding (High 3)

1. Base config is insecurely configured to permit clear text traffic to all domains
2. Domain config is insecurely configured to permit clear text traffic to these domains in scope
3. App can be installed on a vulnerable unpatched Android version 5.0-5.0.2, [minSdk=21]

high	Base config is insecurely configured to permit clear text traffic to all domains	NETWORK
high	Domain config is insecurely configured to permit clear text traffic to these domains in scope	NETWORK
high	App can be installed on a vulnerable unpatched Android version 5.0-5.0.2, [minSdk=21]	MANIFEST

NETWORK SECURITY

1. **Description:** Base config is insecurely configured to permit clear text traffic to all domains

Scope:

*

- 2.**Description:** Domain config is insecurely configured to permit clear text traffic to these domains in scope

Scope:

h.facebook.com

l.facebook.com

l.alpha.facebook.com

lm.facebook.com

l.instagram.com

NO	SCOPE	SEVERITY	DESCRIPTION
1	*	high	Base config is insecurely configured to permit clear text traffic to all domains.
5	h.facebook.com l.facebook.com l.alpha.facebook.com lm.facebook.com l.instagram.com	high	Domain config is insecurely configured to permit clear text traffic to these domains in scope.

3. Description: App can be installed on a vulnerable unpatched Android version 5.0-5.0.2, [minSdk=21]

This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.

NO	ISSUE	SEVERITY	DESCRIPTION	OPTIONS
1	App can be installed on a vulnerable unpatched Android version 5.0-5.0.2, [minSdk=21]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.	

CODE ANALYSIS

issue 5: The App uses an insecure Random Number Generator

CWE-330 | OWASP Top 10: M5 | OWASP MASVS: MSTG-CRYPTO-6

What the vulnerability is:

The application uses Java's `java.util.Random` class, which is a pseudo-random number generator (PRNG) not suitable for security-sensitive operations. Unlike `java.security.SecureRandom`, this class produces predictable sequences if the seed is known or guessed, making it cryptographically weak.

What is specifically wrong:

MobSF identified usage of an insecure random number generator within the application's decompiled source. The affected code relies on `java.util.Random` for generating values that may be used in security-relevant contexts such as token generation, session identifiers, or nonce creation — where unpredictability is a fundamental requirement.

How an attacker could exploit it:

An attacker who knows the algorithm and can observe or deduce the seed value can predict future random values generated by the app. This could allow session token prediction, replay attacks, or bypassing of randomness-dependent security mechanisms within the application.

References:

CWE-330: Use of Insufficiently Random Values | OWASP MASVS: MSTG-CRYPTO-6

issue 6: App can read/write to External Storage. Any App can read data written to External Storage.

warning

CWE-276 | OWASP Top 10: M2 | OWASP MASVS: MSTG-STORAGE-2

File: X/C01V.java

What the vulnerability is:

The application reads from and writes data to external storage (SD card or shared storage). External storage in Android is world-readable by default — any other application installed on the same device with the READ_EXTERNAL_STORAGE permission can access files written there.

What is specifically wrong:

In X/C01V.java, the application performs file I/O operations targeting external storage paths. Storing potentially sensitive user data — such as cached media, session files, or application state — in this location violates the principle of least privilege and exposes that data to unauthorized third-party applications.

How an attacker could exploit it:

A malicious application installed on the same device can read files written by Instagram Lite to external storage. This may expose cached images, downloaded media, or other application data. In a targeted attack, a malicious app could also tamper with files Instagram Lite expects to read, leading to data integrity issues.

References:

CWE-276: Incorrect Default Permissions | OWASP MASVS: MSTG-STORAGE-2

issue 7: MD5 is a weak hash known to have hash collisions.

SEVERITY: warning

CWE-327 | OWASP Top 10: M5 | OWASP MASVS: MSTG-CRYPTO-4

File: X/C0BO.java

What the vulnerability is:

The application uses the MD5 hashing algorithm, which is cryptographically broken. MD5 is known to produce hash collisions — meaning two different inputs can produce the same hash output — making it unsuitable for integrity verification, digital signatures, or any security-critical hashing operation.

What is specifically wrong:

In X/C0BO.java, the application invokes MD5-based hashing. Given that MD5 collision attacks are computationally feasible today, using MD5 in any security context undermines the integrity guarantees the application may rely on. This is compounded by the fact that the application certificate itself is also signed with SHA-1 (identified as a High severity finding in Certificate Analysis).

How an attacker could exploit it:

An attacker could craft a malicious input that produces the same MD5 hash as a legitimate value, bypassing integrity checks. In the context of file verification or token comparison, this could allow substitution attacks where tampered data passes hash validation undetected.

References:

CWE-327: Use of a Broken or Risky Cryptographic Algorithm | OWASP MASVS: MSTG-CRYPTO-4

issue 8: SHA-1 is a weak hash known to have hash collisions. warning CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm

SEVERITY: warning

CWE-327 | OWASP Top 10: M5 | OWASP MASVS: MSTG-CRYPTO-4

File: X/AnonymousClass041.java

What the vulnerability is:

The application uses SHA-1, a hashing algorithm that has been officially deprecated for security use since 2017 following demonstrated collision attacks (SHattered, 2017). SHA-1 no longer provides acceptable collision resistance for cryptographic purposes.

What is specifically wrong:

In X/AnonymousClass041.java, SHA-1 is used for hashing operations. Industry standards and NIST guidelines recommend migrating to SHA-256 or stronger. The continued use of SHA-1 in production code represents a failure to meet current cryptographic standards as defined by OWASP MASVS MSTG-CRYPTO-4.

How an attacker could exploit it:

Using known SHA-1 collision techniques, an attacker could potentially forge hash values used for data verification within the application. If SHA-1 is used for certificate pinning validation, file integrity checks, or authentication tokens, a collision attack could allow the attacker to substitute malicious content that passes the hash check.

References:

CWE-327: Use of a Broken or Risky Cryptographic Algorithm | OWASP MASVS: MSTG-CRYPTO-4

issue 9: IP Address disclosure

SEVERITY: warning

CWE-200 | OWASP MASVS: MSTG-CODE-2

File: X/C00A.java

What the vulnerability is:

The application contains hardcoded IP addresses within its source code. Exposing internal IP addresses in compiled application code constitutes an information disclosure vulnerability, as these addresses can reveal internal network infrastructure details to anyone who reverse-engineers the APK.

What is specifically wrong:

In X/C00A.java, one or more IP addresses are hardcoded as string literals. Since Android APKs can be decompiled with readily available tools (such as JADX or apktool), these IP addresses are easily recoverable by any party who obtains the APK — which is publicly downloadable from the Play Store or the official Instagram Lite website.

How an attacker could exploit it:

A reverse engineer can decompile the APK, extract the hardcoded IP addresses from X/C00A.java, and use them to map the application's backend infrastructure. This information can facilitate targeted network attacks, server enumeration, or denial-of-service attempts against the exposed endpoints.

References:

CWE-200: Information Exposure | OWASP MASVS: MSTG-CODE-2

NO	ISSUE	SEVERITY	STANDARDS	FILES	OPTIONS
2	The App uses an insecure Random Number Generator.	warning	CWE: CWE-330: Use of Insufficiently Random Values OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-6	Show Files	
3	IP Address disclosure	warning	CWE: CWE-200: Information Exposure OWASP MASVS: MSTG-CODE-2	X/C00A.java	
5	MD5 is a weak hash known to have hash collisions.	warning	CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-4	X/C0B0.java	
6	SHA-1 is a weak hash known to have hash collisions.	warning	CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-4	X/AnonymousClass041.java	
7	App can read/write to External Storage. Any App can read data written to External Storage.	warning	CWE: CWE-276: Incorrect Default Permissions OWASP Top 10: M2: Insecure Data Storage OWASP MASVS: MSTG-STORAGE-2	X/C01V.java	

TRACKERS

Description:

This application has no privacy trackers

This application does not include any user or device trackers. Unable to find trackers during static analysis.

secure This application has no privacy trackers

TRACKERS

This application does not include any user or device trackers. Unable to find trackers during static analysis.

4.Top Malware Permissions 16/25

android.permission.ACCESS_COARSE_LOCATION,
 android.permission.ACCESS_FINE_LOCATION,
 android.permission.ACCESS_NETWORK_STATE,
 android.permission.ACCESS_WIFI_STATE, android.permission.CAMERA,
 android.permission.GET_TASKS, android.permission.INTERNET,
 android.permission.READ_CONTACTS, android.permission.GET_ACCOUNTS,
 android.permission.READ_PHONE_STATE,
 android.permission.RECEIVE_BOOT_COMPLETED,

android.permission.RECORD_AUDIO,
android.permission.SYSTEM_ALERT_WINDOW, android.permission.VIBRATE,
android.permission.WAKE_LOCK,
android.permission.WRITE_EXTERNAL_STORAGE

5. Malicious analysis from VirusTotal (virustotal.com) :

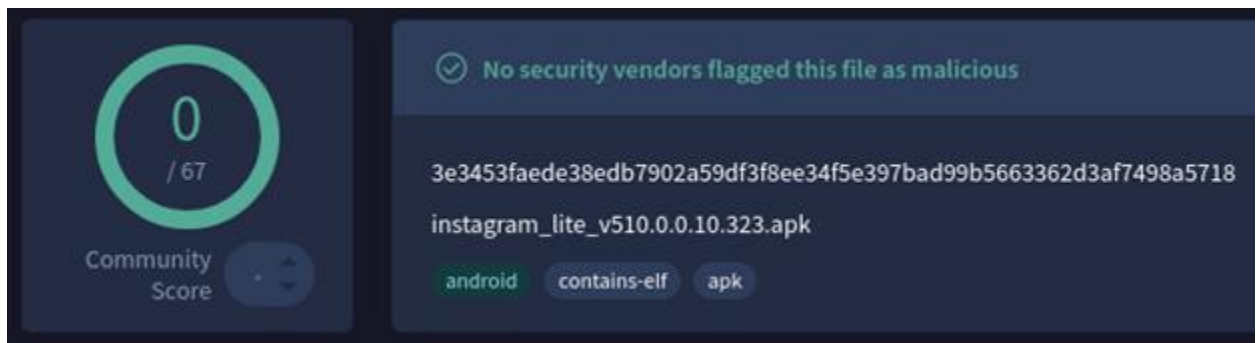
instagram_lite_v510.0.0.10.323.apk

SHA256

3e3453faede38edb7902a59df3f8ee34f5e397bad99b5663362d3af7498a5718

Community Score 0/67

No security vendors flagged this file as malicious



Though, Permissions requested by the file being studied. To maintain security for the system and users, Android requires apps to request permission before the apps can use certain system data and features.

Description of Permission:

android.permission.READ_PHONE_NUMBERS

This app can read all calendar events stored on your phone and share or save your calendar data.

android.permission.READ_CALENDAR

This app can get your approximate location from location services while the app is in use. Location services for your device must be turned on for the app to get location.

android.permission.ACCESS_COARSE_LOCATION

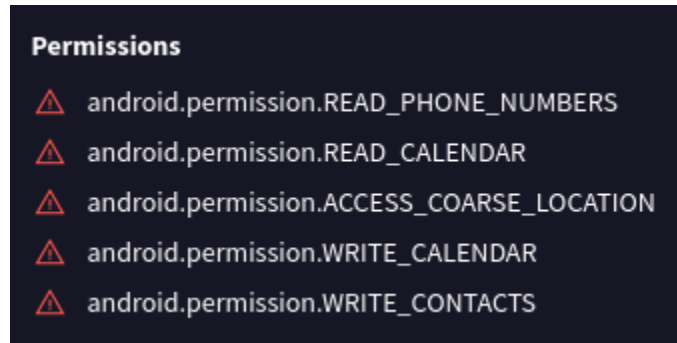
This app can add, remove, or change calendar events on your phone. This app can send messages that may appear to come from calendar owners, or change events without notifying their owners.

android.permission.WRITE_CALENDAR

Allows the app to modify the data about your contacts stored on your phone. This permission allows apps to delete contact data.

android.permission.WRITE_CONTACTS

& more



6.Trackers & Permission Analysis By Exodus Privacy :

Instagram Lite

Version 510.0.0.10.323

Source: <https://www.instagram.com/web/lite/>

Trackers - 0

Permission - 61



Instagram Lite

0 trackers

61 permissions

Version 510.0.0.10.323 - [see other versions](#)

Source: Google Play

Report created on May 8, 2026, 3:24 a.m.

[See on Google Play >](#)

0 trackers

We have not found **code signature** of any tracker we know in the application.

The application could contain tracker(s) we do not know yet.

A tracker is a piece of software meant to collect data about you or your usages. [Learn more...](#)

61 permissions

We have found the following permissions in the application:

 **! ACCESS_COARSE_LOCATION**
access approximate location only in the foreground

 **! ACCESS_FINE_LOCATION**
access precise location only in the foreground

ACCESS_NETWORK_STATE
view network connections

ACCESS_WIFI_STATE
view Wi-Fi connections

AUTHENTICATE_ACCOUNTS

BATTERY_STATS

BIND_NOTIFICATION_LISTENER_SERVICE

BLUETOOTH
pair with Bluetooth devices

BROADCAST_STICKY
send sticky broadcast

 **! CALL_PHONE**
directly call phone numbers

 **! CAMERA**
take pictures and videos

CHANGE_NETWORK_STATE
change network connectivity

CHANGE_WIFI_STATE
connect and disconnect from Wi-Fi

CREDENTIAL_MANAGER_SET_ALLOWED_PROVIDERS

FOREGROUND_SERVICE
run foreground service

FOREGROUND_SERVICE_DATA_SYNC

FOREGROUND_SERVICE_MEDIA_PROJECTION

FOREGROUND_SERVICE_MICROPHONE

 **! GET_ACCOUNTS**
find accounts on the device

GET_TASKS
retrieve running apps

INTERNET
have full network access

MANAGE_ACCOUNTS

POST_NOTIFICATIONS

 **! READ_CALENDAR**
Read calendar events and details

 **! READ_CONTACTS**
read your contacts

READ_MEDIA_VISUAL_USER_SELECTED

 **! READ_PHONE_NUMBERS**
read phone numbers

 **! READ_PHONE_STATE**
read phone status and identity

READ_PROFILE

RECEIVE_BOOT_COMPLETED
run at startup

 **! RECORD_AUDIO**
record audio

REORDER_TASKS
reorder running apps

RUN_USER_INITIATED_JOBS

! SYSTEM_ALERT_WINDOW
This app can appear on top of other apps

USE_FULL_SCREEN_INTENT
display notifications as full screen activities on a locked device

VIBRATE
control vibration

WAKE_LOCK
prevent phone from sleeping

 **! WRITE_CALENDAR**
add or modify calendar events and send email to guests without owners' knowledge

 **! WRITE_CONTACTS**
modify your contacts

 **! WRITE_EXTERNAL_STORAGE**
modify or delete the contents of your shared storage

INSTALL_SHORTCUT
install shortcuts

UNINSTALL_SHORTCUT
uninstall shortcuts

ACCESS

ACCESS

ACCESS

ACCESS

ACCESS

Chapter 4 — Other Activities

1. Checklist Analysis

A security checklist review was conducted based on OWASP Mobile Security Guidelines (OWASP MASVS) to systematically verify Instagram Lite's security posture:

Security Area	Check	Status
Certificate Strength	SHA-256 or stronger used?	FAIL
Network Security	HTTPS enforced for all domains?	FAIL
Data Storage	No world-writable files/preferences?	FAIL
Cryptography	Secure cipher modes used?	FAIL
WebView Security	Remote debugging disabled?	N/A
Malware Permissions	No top malware permissions used?	FAIL
Backup Settings	allowBackup disabled?	N/A
VirusTotal Clean	No vendor flagged as malicious?	PASS
Tracker Count	Minimal trackers?	PASS

This checklist confirms that Instagram Lite fails in **6** out of **9** key security areas, reinforcing the medium-risk rating of **53/100** assigned by MobSF.

- 2. Other Activities could be done :** Additional investigative or validation steps can also be performed beyond this process — such as **Dynamic Testing, API Endpoint Analysis, and attempts at Manual Penetration Testing**; Completing these steps would make the security assessment of the APK or Android application more precise, comprehensive, and comparatively much stronger, along with providing more effective remediation measures.

Chapter 5— Conclusion

Inconsistency acknowledges:

MobSF detected **0** trackers during static analysis, while Exodus Privacy's database shows **0** trackers for this app version — indicating additional trackers may be identified through dynamic analysis.

The static analysis of Instagram Lite v510.0.0.10.323 using MobSF revealed a security score of 53/100, indicating Medium risk with 3 high-severity findings. VirusTotal confirmed no malware detection **0/67**, while Exodus Privacy identified **0** trackers and **61** permissions — raising significant user privacy concerns. Overall, despite its widespread use, Instagram Lite requires substantial security improvements, particularly in **network security (cleartext traffic permitted to all domains and specific Instagram domains)**, **22 dangerous permissions including high-risk ones such as CAMERA, RECORD_AUDIO, SYSTEM_ALERT_WINDOW, READ_CONTACTS, and READ_PHONE_STATE**, use of a weak SHA1 with RSA certificate, insecure external storage access, unprotected exported components (**21 activities, 15 receivers, 8 providers**), support for vulnerable Android 5.0 (minSdk=21), and **61 total permissions** flagged by Exodus Privacy.